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作业9.

思考题

1. (1). $\int_0^1 dy \int_0^1 \frac{x^2-y^2}{(x^2+y^2)^2} dx$

$= \int_0^1 dy \left(-\frac{x}{x^2+y^2}\Big|_0^1\right)$

$= \int_0^1 \frac{1}{1+y^2} dy = \arctan y\Big|_0^1 = -\frac{\pi}{4}$

(2). $\int_0^1 dx \int_0^1 \frac{x^2-y^2}{(x^2+y^2)^2} dy$

$= \int_0^1 dx \left(\frac{y}{x^2+y^2}\Big|_0^1\right)$

$= \int_0^1 \frac{1}{x^2+1} dx = \arctan x\Big|_0^1 = \frac{\pi}{4}$

(3). 依比尼定理要求绝对值积分 $\iint_{[0,1]^2} \left|\frac{x^2-y^2}{(x^2+y^2)^2}\right| dxdy$

不发散，即 $\left|\frac{x^2-y^2}{(x^2+y^2)^2}\right|$ 在 $[0,1]^2$ 上须绝对可积，才可换序

而其不满足绝对可积条件，故不可换序.

(4). 令 $x = r\cos\theta$, $y = r\sin\theta$, $\theta \in [0, \frac{\pi}{4}]$

$\therefore x^2-y^2 = r^2\cos 2\theta$, $x^2+y^2 = r^2$.

$\therefore \left|\frac{x^2-y^2}{(x^2+y^2)^2}\right| = \frac{\cos 2\theta}{r^2}$, $dxdy = r\,dr\,d\theta$.

$\therefore$ 构造积分 I_1: $\left|\frac{x^2-y^2}{(x^2+y^2)^2}\right| dxdy = \left|\frac{\cos 2\theta}{r}\right| dr\,d\theta$. 单位圆内

则 $I_1 = \int_0^1\int_0^{\frac{\pi}{4}} \frac{|\cos 2\theta|}{r} d\theta\, dr = \int_0^{\frac{\pi}{4}} |\cos 2\theta| d\theta \int_0^1 \frac{1}{r} dr = \infty$，发散.

又 $\because \left|\frac{x^2-y^2}{(x^2+y^2)^2}\right| > 0$ $\therefore \iint_{[0,1]^2} \left|\frac{x^2-y^2}{(x^2+y^2)^2}\right| dxdy > I_1$ $\therefore$ 此积分也发散

即: $\iint_{[0,1]^2} \left|\frac{x^2-y^2}{(x^2+y^2)^2}\right| dxdy = \infty$

书后题：

9.1 (3). $\iint\limits_D \sqrt{19-y^2}\,d\sigma$

$= \int_0^4 dx \int_0^3 \sqrt{19-y^2}\,dy$

$= \frac{9}{4}\pi \cdot 4 = 9\pi$,

2. (3). $\rho(r) = Cr$ 即 $\rho(x,y,z) = C\sqrt{x^2+y^2+z^2}$.

$\therefore m = \int_V \rho\, dV$

$= C \iiint\limits_{x^2+y^2+z^2 \le R^2} \sqrt{x^2+y^2+z^2}\, dV$.

3. (3) $\because |x+y| \ge |\sin(x+y)|$

$\therefore (x+y)^2 \ge (\sin^2(x+y))$

$\therefore I_2 \ge I_1$ $\because$ D的面积大于0.

$\therefore I_2 > I_1$.

4. (4) $\because 0 \le x^2+y^2 \le \frac{1}{4}$,

$\therefore 1 \le e^{x^2+y^2} \le e^{\frac{1}{4}}$

$\therefore \frac{\pi}{4} < I < \frac{\pi}{4} e^{\frac{1}{4}}$.

5. $f(x,y) = f(x_0,y_0) + O[(x-x_0)+(y-y_0)]$

$\therefore$ 原式 $= \frac{1}{\pi r^2} \iint\limits_{D_r} f(x_0,y_0)\, d\sigma = f(x_0,y_0)$.

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6.6. $\because f(x,y) \geq 0$.

$\therefore$ 区域D面积为A.

$\therefore$ 若 $f(x,y) \neq 0$.

$\therefore$ 设 $f(x,y) \geq \varepsilon$, ε 为任意小正数

$\therefore \iint\limits_D f(x,y)\,d\sigma \geq \varepsilon A > 0$. $\therefore$ 不成立.

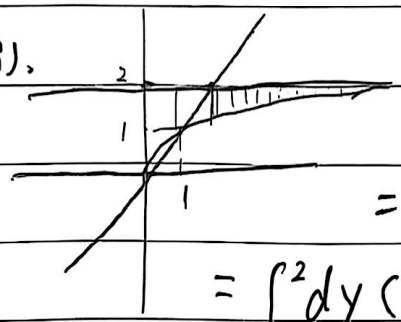
$\therefore f(x,y) \equiv 0$.

7.(4). $\iint\limits_D f(x,y)\,d\sigma$

$= \int_{-1}^{1} dx \int_{x^3}^{1} f(x,y)\,dy$

$= \int_{-1}^{1} dy \int_{-1}^{\sqrt[3]{y}} f(x,y)\,dx$.

8.(8).



$\iint\limits_D \sin\left(\frac{x}{y}\right) dx\,dy$

$= \int_1^2 dy \int_y^{y^3} \sin\left(\frac{x}{y}\right) dx$.

$= \int_1^2 dy\,(y\cos 1 - y\cos y^2)$

$= \frac{3}{2}\cos 1 - \int_1^2 y\cos y^2\,dy$.

$= \frac{3}{2}\cos 1 - \frac{1}{2}(\sin 4 - \sin 1)$

$= \frac{1}{2}(3\cos 1 + \sin 1 - \sin 4)$

9. $\iint\limits_D f(x)g(y)\,dx\,dy$

$= \int_a^b dx \int_c^d f(x)g(y)\,dy.$

$= \int_a^b dx \left(f(x)\int_c^d g(y)\,dy\right)$

$= \int_a^b f(x)\,dx \int_c^d g(y)\,dy.$

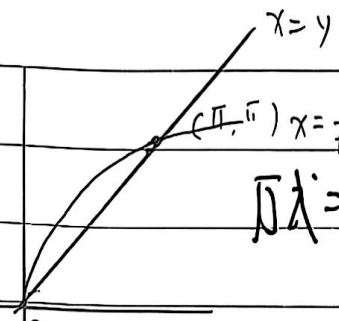
10. $y=\sqrt{4-x^2} \Rightarrow x=\sqrt{4-y^2}.$

$y=\sqrt{2x-x^2} \Rightarrow (x-1)^2+y^2=1$

$\therefore \text{原式} = \int_1^2 dy \int_0^{\sqrt{4-y^2}} f(x,y)\,dx + \int_0^1 dy \int_0^{1-\sqrt{1-y^2}} f(x,y)\,dx$

$+ \int_0^1 dy \int_{1+\sqrt{1-y^2}}^{\sqrt{4-y^2}} f(x,y)\,dx$

11. (6)



$\text{原式} = \int_0^{\pi} dy \int_{\frac{1}{\pi}y^2}^{y} \frac{\sin y}{y}\,dx$

$= \int_0^{\pi} \left(\frac{\sin y}{y}\right)\left(y-\frac{1}{\pi}y^2\right)dy.$

$= \int_0^{\pi} \left(\sin y - \frac{1}{\pi} y \sin y\right) dy$

$= 2 + \frac{1}{\pi}\left(y\cos y\Big|_0^{\pi} - \int_0^{\pi} \cos y\,dy\right) = 1$

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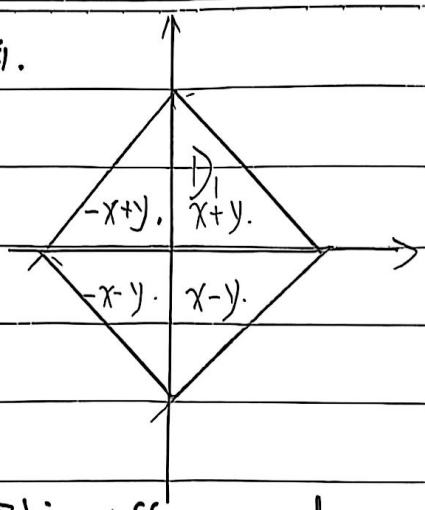
12. (4).



$\therefore \iint\limits_{D} = 4\iint\limits_{D_1}(x+y)\,d\sigma$

$= 4\int_0^1 dx \int_0^{1-x}(x+y)\,dy.$

$= 4\int_0^1 \left[x(1-x)+\frac{1}{2}(1-x)^2\right]dx$

$= 2\int_0^1 (1-x^2)\,dx = \frac{4}{3}.$

23 ①. $\iint\limits_{D} = \int_0^2 dx \int_{-3}^0 dy \int_{-1}^1 (x^2+y^2)\,dz$

$= \int_0^2 dx \int_{-3}^0 2x^2\,dy$

$= \int_0^2 6x^2\,dx = 2x^3\Big|_0^2 = 16$

②. $\iint\limits_{D} = \int_0^2 dx \int_{-1}^1 dz \int_{-3}^0 (x^2+y^2)\,dy$

$= \int_0^2 dx \int_{-1}^1 (3x^2-\frac{9}{2}z)\,dz.$

$= \int_0^2 (6x^2)\,dx = 16$

③. $\iint\limits_{D} = \int_{-3}^0 dy \int_0^2 dx \int_{-1}^1 (x^2+y^2)\,dz.$

$= \int_{-3}^0 dy \int_0^2 2x^2\,dx = \int_{-3}^0 \frac{16}{3}\,dy = 16.$

24. 由 $\begin{cases} z=xy \\ z=0 \end{cases} \Rightarrow x=0$ 或 $y=0$.

又 $\because x+y-1=0$.

$\therefore$ 在 xOy 平面内：投影区域为 $\{(x,y) \mid x\geq 0,\ y\geq 0,\ x+y\leq 1\}$

$\therefore \iiint\limits_{\Omega} f(x,y,z)\,dV = \int_0^{xy} dz \int_0^1 dx \int_0^{1-x} f(x,y,z)\,dy$.

29.(8). 令 $\begin{cases} x=4 \\ x=4y^2+4z^2 \end{cases}$ $\therefore y^2+z^2=1$ 　　即区域：$\begin{cases} y^2+z^2\leq \frac{1}{4}x \\ 0\leq x\leq 4 \end{cases}$

$\therefore \iiint\limits_{\Omega} x\,dx\,dy\,dz$

$= \int_{-1}^{1} dz \int_{-\sqrt{1-z^2}}^{\sqrt{1-z^2}} dy \int_{4(y^2+z^2)}^{4} x\,dx$

$= \int_{-1}^{1} dz \int_{-\sqrt{1-z^2}}^{\sqrt{1-z^2}} (8-8(y^2+z^2)^2)\,dy$. 令 $r^2=y^2+z^2$

$= \iint\limits_{D'} (8-8(y^2+z^2)^2)\,d\sigma$ 　　$\because d\sigma = r\,dr\,d\theta$

$= 8\iint\limits_{D} d\sigma - 8\int_0^1 r^5\,dr \int_0^{2\pi} d\theta$

$= \frac{16}{3}\pi$.